

AKSHAR AI

SUMMER INTERNSHIP REPORT

WEEK 2 | 30 MAY 2026 – 05 JUNE 2026

Project Title:

Chest X-Ray Pneumonia Detection and Analysis System Using Deep Learning

Name	Yashas Raj R
Enrollment No.	24162121033
Branch	BDA (B.Tech – Computer Science and Engineering)
Institute	ICT Ganpat University, Mehsana, Gujarat
Organisation	Akshar AI Ltd
Report Period	30 May 2026 – 05 JUNE 2026 (Week 2)
Email	24162121033@gnu.ac.in
Project Title	Pneumonia Detection System
Google Drive Link	WEEK-02

1. Introduction

Pneumonia is one of the leading infectious causes of mortality worldwide, disproportionately affecting children under five years of age and elderly populations. The disease causes the alveoli in the lungs to fill with fluid and inflammatory cells, which severely impairs the body's ability to exchange oxygen and carbon dioxide. Early and accurate detection is, therefore, of paramount clinical importance.

Chest X-ray (CXR) imaging is the most widely used frontline diagnostic tool for pneumonia owing to its accessibility, low cost, and fast turnaround. However, the manual interpretation of chest radiographs is an inherently complex task that requires trained radiologists to identify subtle parenchymal opacities, consolidation patterns, and bilateral infiltrates — patterns that can be easily missed under conditions of high workload or fatigue.

Artificial Intelligence, and Deep Learning in particular, has emerged as a transformative force in medical imaging. Convolutional Neural Networks (CNNs) have demonstrated the ability to learn hierarchical spatial representations from pixel data and match or exceed radiologist-level performance on binary classification tasks such as distinguishing normal from pneumonia-affected chest X-rays. This capability makes AI-driven systems highly relevant in resource-limited healthcare environments where specialist radiologists may not always be available.

Beyond the deep learning model itself, the usability and accessibility of the system that wraps the model is equally critical. A clinically powerful model loses much of its value if it is delivered through a poorly designed interface. Medical professionals and patients alike benefit from applications that are intuitive, trustworthy, and visually reassuring. Research in healthcare UX consistently shows that interface design affects user confidence, accuracy of interaction, and overall adoption.

Week-02 of this internship was therefore dedicated to three parallel streams of work: reviewing and documenting the achievements of Week-01, deepening theoretical and practical knowledge of React.js for frontend development, and conducting rigorous research into color psychology and healthcare-focused UI/UX design principles. The knowledge and design decisions developed during this week will form the foundation upon which the full web-based Pneumonia Detection System interface will be built.

2. Week-02 Objectives

The objectives set out for Week-02 were designed to bridge the gap between the data-focused activities of Week-01 and the frontend development work scheduled for the coming weeks. They span documentation, learning, and design research:

- Review and evaluate the progress and outputs achieved during Week-01, including dataset exploration and EDA deliverables.
- Prepare a comprehensive Week-01 internship report with day-wise documentation of all technical activities.
- Revise and reinforce core React.js concepts in preparation for frontend implementation.
- Study modern frontend development practices and component-based architecture principles.
- Research color psychology with a focus on how color choices affect human emotion, trust, and readability.
- Analyze healthcare-specific UI/UX principles, with particular attention to accessibility and emotional design.
- Plan the overall structure of the homepage and authentication system (login/registration) for the Pneumonia Detection System.
- Prepare a structured readiness plan for the upcoming frontend implementation sprint.

3. Day-Wise Work Summary

Day 1 — 30 May 2026 (Saturday) | Project Review and Planning

The week commenced with a structured weekly project review meeting with the Akshar AI project team. This session served both as a retrospective on Week-01 deliverables and as a planning workshop for the activities ahead.

During the review, the team discussed the key achievements from Week-01: the successful download and organisation of the chest X-ray dataset from Kaggle into train, validation, and test directories; the dataset validation process using Python's `os`, `pathlib`, and `glob` libraries; the completion of a thorough Exploratory Data Analysis including class distribution analysis, image resolution scatter plots, grayscale pixel intensity histograms, and a structural artifact audit; and, finally, the deployment of a React-based visualisation application that rendered all EDA outputs in an interactive, browser-accessible format.

From this review, two clear priorities emerged for the week ahead. First, the team directed that a detailed Week-01 report be written and submitted, capturing each day's activities with technical depth and professional formatting. Second, attention was to shift toward the design and development of the main web application interface — specifically the homepage, login page, and registration page. To do this effectively, I was asked to research color psychology and emotional design in healthcare contexts, and to revise React fundamentals in preparation for implementation.

I found the review session particularly valuable for situating my individual work within the larger project architecture. Understanding that the EDA work feeds directly into model development, and that the frontend application must eventually present model predictions in a clinically responsible and user-friendly way, helped me appreciate why the design research work is as critical as the technical modelling.

Day 2 — 01 June 2026 (Monday) | React Revision and Frontend Preparation

With the frontend development sprint approaching, I dedicated this day to a thorough revision of React.js fundamentals. I worked through the YouTube tutorial 'Learn React JS — Full Beginner's Tutorial & Practice Projects', making detailed notes and referring to the associated GitHub repository of practice examples.

The revision session covered several foundational concepts. In terms of Components and JSX, I reviewed how React components serve as the primary building blocks of any user interface, and how JSX allows developers to write HTML-like syntax directly within JavaScript, which is then transpiled into `React.createElement()` calls. I re-familiarised myself with the distinction between functional components and class components, and with the move toward functional components as the modern best practice.

For Props and State, I revisited how props enable parent components to pass data down to child components, and how the useState hook manages mutable data within a component. Understanding this distinction is essential for building the Pneumonia Detection System's UI, where components such as the image upload widget, the results display panel, and the navigation bar will each maintain their own internal states while receiving configuration data from parent components.

On the topic of Event Handling, I reviewed how React wraps DOM events in SyntheticEvent objects for cross-browser consistency, and how handlers like onClick, onChange, and onSubmit are used to respond to user interactions. I also revised the useEffect hook for managing side effects such as data fetching and DOM manipulation after a component renders.

The session reinforced my appreciation for React's component-based architecture as a scalable approach to building complex healthcare interfaces. By composing the UI from small, independently testable components — such as a NavBar, HeroSection, FeatureCard, LoginForm, and RegisterForm — the application becomes far easier to maintain and extend as new features are added over the course of the internship.

Day 3 — 02 June 2026 (Tuesday) | Research on Color Psychology and UI/UX Design

This day was devoted to understanding the psychological dimensions of color in digital interfaces, with a specific focus on healthcare applications. I studied two research papers and watched two YouTube tutorials on color theory and UI/UX design.

From the paper 'Color and Psychological Functioning: A Review of Theoretical and Empirical Work', I learned that color perception is deeply intertwined with human psychological states. Colors influence arousal levels, emotional valence, cognitive performance, and interpersonal trust. The paper reviewed theories ranging from Elliot and Maier's Color-in-Context Model — which posits that color meanings are context-dependent — to evolutionary perspectives on why certain hues evoke specific responses. For a medical application like the Pneumonia Detection System, where users may be anxious patients or time-pressured clinicians, these insights are highly actionable.

The second paper, 'The Effect of Ambient Illumination and Text Color on Visual Fatigue', examined how color contrasts between text and background affect eye strain during extended screen use. Key findings indicated that high-contrast but not overly harsh color combinations — such as dark navy text on off-white backgrounds — significantly reduce visual fatigue compared to pure black on white. This finding is directly relevant to the design of report display pages and data dashboards within the Pneumonia Detection System.

From the YouTube tutorials on color theory and design, I explored the practical application of color wheels, complementary color schemes, analogous palettes, and the psychology of specific hues. Blue and its variants consistently emerged as the dominant choice for healthcare applications because they evoke calmness,

professionalism, and institutional trust — qualities that are essential when patients are interacting with a system that provides medical information. Green supports associations with health, vitality, and positive outcomes, making it a strong accent color for success indicators. Warm reds and oranges, by contrast, should be used sparingly and only for alert or warning states.

I also discovered the importance of accessibility in color design. Approximately 8% of males and 0.5% of females experience some form of color vision deficiency. Designing the system with sufficient luminance contrast and avoiding red-green combinations ensures that the application is usable by a broader range of healthcare professionals and patients.

Day 4 — 03 June 2026 (Wednesday) | Week-01 Report Documentation

This day was entirely devoted to writing, structuring, and finalising the Week-01 internship report. The report covered the five working days from 25 May to 29 May 2026, during which the team focused on dataset exploration and EDA.

The documentation process began with a full review of all notes, code outputs, and visualisations produced during Week-01. I organised the content day-wise, ensuring each day's entry clearly described the objectives pursued, the specific tasks executed, the tools and libraries used, the observations made, and the outcomes achieved.

Documenting the dataset collection phase required explaining the use of Python's `os` and `pathlib` modules to traverse the Kaggle directory structure, the validation checks run to identify corrupted or zero-byte image files, and the metadata extraction process using `OpenCV` and `PIL`. For the EDA section, I described the visualisation grids created with `Matplotlib`, the class distribution bar plots generated using `Seaborn`, the resolution scatter plots, the pixel intensity histograms, and the structural artifact audit designed to identify non-biological markers such as orientation labels and surgical hardware.

Writing this report was a valuable exercise in itself. Articulating technical processes in clear, precise language requires a deep understanding of what was done and why. The act of writing also revealed a few areas where my Week-01 documentation could have been more thorough — for example, recording the exact counts of corrupted files found and noting the specific augmentation strategies considered during the EDA phase. These are lessons I have carried forward into the documentation of Week-02.

The final report was organised, formatted, and saved to Google Drive for review by both the academic supervisor and the company mentor.

Day 5 — 04 June 2026 (Thursday) | Healthcare UI Research and Frontend Design Planning

With the color psychology research from Day 3 as a foundation, I shifted focus to analysing modern healthcare websites and medical dashboard interfaces to extract design patterns relevant to the Pneumonia Detection System.

I reviewed the UI/UX approaches of several leading healthcare platforms, noting common patterns in typography, layout, navigation, and interaction design. Clean, sans-serif typefaces dominate healthcare UIs because they are more legible at small sizes and project a sense of precision. Generous white space helps users focus on critical information without visual clutter. Navigation menus are consistently minimal and logically grouped — patients should never have to search more than two clicks to find the function they need.

For the homepage of the Pneumonia Detection System, I planned a hero section with a concise value proposition, a features overview section highlighting the AI model's capabilities, and a clear call-to-action directing users to the upload interface or the login page. The colour palette will be anchored in the blue family informed by the color psychology research, with a white main background, navy primary text, blue accent buttons, and green success indicators.

For the authentication system, I planned a login page with email and password fields, a 'Remember Me' checkbox, and a link to the registration page. The registration page will request a username, email, password with confirmation, and a role selector to distinguish between patients and healthcare professionals. I also planned error state messaging, loading indicator states, and field validation feedback — all critical elements for a trustworthy and frustration-free authentication experience.

Accessibility considerations were central to the planning. I noted the need for ARIA labels on all form inputs, keyboard-navigable focus states, and sufficient color contrast ratios following WCAG 2.1 AA guidelines. Mobile responsiveness was also planned from the outset, with a mobile-first approach ensuring that the application functions equally well on smartphones as on desktop workstations — an important consideration given the increasing use of mobile devices by healthcare workers in clinical settings.

Day 6 — 05 June 2026 (Friday) | Consolidation of Learning and Development Preparation

The final day of Week-02 was a structured review and consolidation session. I revisited the week's key learning outputs — the React revision notes, the color psychology research summaries, the healthcare UI analysis findings, and the Week-01 report — and synthesised them into a coherent development readiness document.

In reviewing the React revision, I confirmed that I am comfortable with the core concepts needed to build the initial frontend components: functional components, JSX syntax, useState and useEffect hooks, event handlers, and React Router for navigation between pages. I identified a few areas requiring further practice

— specifically, form validation logic and conditional rendering patterns — which I plan to address in a short focused exercise before beginning the implementation sprint.

In reviewing the color psychology and UI research, I finalised the preliminary color palette for the Pneumonia Detection System: Navy Blue (#1F4E79) for primary headings and navigation backgrounds, Medium Blue (#2E75B6) for interactive elements and highlights, Soft Teal (#17A589) for success and positive-result indicators, Light Gray (#F4F6F9) for page backgrounds, and White (#FFFFFF) for card surfaces. This palette is grounded in the psychological research reviewed earlier in the week and aligns with industry standards for healthcare digital products.

By the end of the day, I felt a genuine sense of readiness for the implementation phase. The research, planning, and documentation work of Week-02 has transformed what could have been a week of abstract preparation into a structured, evidence-based foundation from which the frontend development can proceed with clarity and confidence.

4. Research Activities

4A. React.js Learning

My React revision this week was structured around understanding how modern React applications are architected and how each core concept contributes to building scalable, maintainable user interfaces. Below is a summary of the key topics studied:

Components and JSX

React components are the fundamental units of a React application. Each component encapsulates a piece of the user interface along with its own logic and state. JSX — JavaScript XML — allows developers to describe UI structures using syntax that closely resembles HTML but compiles to standard JavaScript function calls. I practised creating both presentational components (which simply render data they receive via props) and container components (which manage state and pass data down to presentational children).

Props and State Management

Props are the mechanism by which parent components communicate data to their children, while state represents data that can change over time within a component. Understanding the unidirectional data flow in React — where data flows from parent to child, and events bubble up from child to parent via callback props — is fundamental to designing the layout of the Pneumonia Detection System's component hierarchy.

Hooks: useState and useEffect

The useState hook allows functional components to maintain mutable data without the need for class syntax. The useEffect hook manages side effects — such as fetching prediction results from the backend API, updating the document title, or subscribing to WebSocket events — and provides a clean-up mechanism to prevent memory leaks. These two hooks together cover the vast majority of stateful behaviour needed in the Pneumonia Detection System frontend.

Event Handling

React wraps native browser events in a cross-browser consistent SyntheticEvent interface. I revised patterns for handling onClick events on buttons, onChange events on form inputs, and onSubmit events on forms. Proper event handling is essential for the file upload functionality and the authentication forms that will be implemented in Week-03.

Component-Based Architecture and Reusability

One of React's greatest strengths is the ease with which components can be reused across an application. A well-designed Button component, for example, can be parameterised with different labels, colours, sizes, and

onClick handlers and used consistently across every page of the Pneumonia Detection System — ensuring a coherent visual language and reducing the volume of duplicated code.

4B. Color Psychology Research

Human Perception of Color

Color is perceived when light of specific wavelengths stimulates the photoreceptor cells in the human retina. Beyond the purely physiological process, however, color perception is heavily modulated by cultural context, personal experience, and situational framing. The same shade of red may signify urgency, danger, passion, or celebration depending on the context in which it is encountered.

Color-Emotion Relationships

Research consistently identifies a set of core color-emotion associations that transcend cultural variation to a significant degree. Blue is associated with calmness, competence, and trust. Green signals health, safety, and permission. Red triggers alertness, urgency, and attention. White conveys cleanliness, clarity, and openness. Black and dark navy project authority and professionalism. These associations have direct implications for the design of a medical application.

Trust and Reliability in Healthcare Interfaces

Trust is perhaps the most critical quality a healthcare application must convey. Users interacting with a system that provides medical information are in a vulnerable psychological state and are highly sensitive to visual cues of professionalism. Consistent use of blue-dominant palettes, restrained typography, and high contrast ratios signals institutional reliability. Conversely, overly bright, saturated, or playful color schemes may undermine user confidence in a medical context.

Readability, Eye Comfort, and Accessibility

Extended screen interaction in clinical environments places demands on readability that differ from those of consumer applications. High contrast between text and background reduces eye strain; however, pure black (#000000) on pure white (#FFFFFF) can create excessive luminance contrast that contributes to fatigue over long sessions. Research recommends slightly softened combinations — such as near-black on off-white or medium navy on light gray — for extended reading contexts. Additionally, designers must ensure that color is not the sole carrier of information, given the prevalence of color vision deficiencies in the population.

4C. UI/UX Research for Healthcare Applications

User-Centered Design Principles

The foundational principle of user-centered design is that every design decision must be evaluated from the perspective of the intended user. For the Pneumonia Detection System, the user base includes clinicians who

need fast, accurate, and clearly labelled results, and patients who may have limited health literacy and who need reassuring, simple language and interfaces.

Accessibility and WCAG Compliance

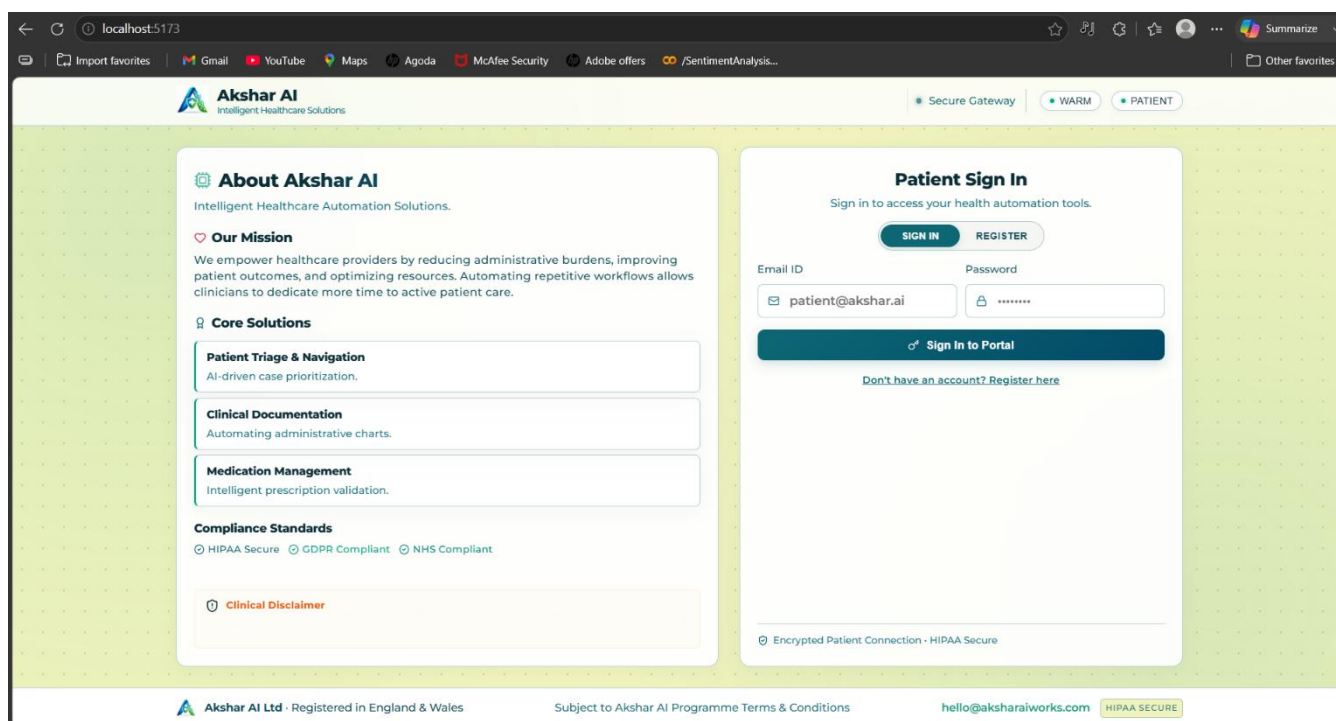
The Web Content Accessibility Guidelines (WCAG) 2.1 define a set of standards for making digital content accessible to users with disabilities. At the AA compliance level, all text must achieve a minimum contrast ratio of 4.5:1 against its background. All interactive elements must be reachable and operable via keyboard navigation. Images must carry meaningful alternative text descriptions. These requirements will be applied throughout the Pneumonia Detection System's interface.

Navigation and Visual Hierarchy

Effective navigation design ensures that users can orient themselves within the application at any point and reach their desired function with minimal effort. For the Pneumonia Detection System, a persistent top navigation bar with clear labels — Home, Upload, Results, About — provides the global navigation structure. Visual hierarchy within individual pages is established through typographic scale (larger headings, smaller body text), spatial grouping, and the strategic use of color to direct attention to the most important elements.

Responsive Design for Multi-Device Access

Healthcare professionals increasingly access clinical tools on mobile devices. A mobile-first design approach — where the layout is designed for small screens first and then expanded for larger ones — ensures that the Pneumonia Detection System is accessible on smartphones, tablets, and desktop workstations alike. React's component model, combined with CSS media queries and modern layout systems such as Flexbox and CSS Grid, supports this responsive approach naturally.



5. Tools and Technologies Used

The following table summarises the tools and technologies engaged during Week-02, along with a description of their specific purpose within the internship activities:

Tool / Technology	Purpose	Description
React.js	Frontend Development	A declarative JavaScript library for building component-based user interfaces. Revised and studied for the Pneumonia Detection System frontend.
JavaScript (ES6+)	Application Logic	The core scripting language of modern web applications. Used for component logic, event handling, and API integration.
HTML5	Page Structure	The standard markup language for web pages. Provides semantic structure for components such as navigation, forms, and content sections.
CSS3	Styling and Layout	Used for visual styling, responsive layouts (Flexbox, Grid), animations, and media queries for mobile responsiveness.
Git	Version Control	Command-line version control system for tracking code changes, managing feature branches, and maintaining project history.
GitHub	Collaboration	Remote repository hosting service used for storing, sharing, and collaborating on project code with the Akshar AI team.
VS Code	Development Environment	The primary code editor used throughout the internship. Extensions for React, ESLint, Prettier, and GitLens enhance productivity.
Google Scholar	Research Papers	Academic paper discovery platform. Used to find peer-reviewed research on color psychology and healthcare UX design.
YouTube	Learning Resources	Video-based learning platform. Used for React tutorial revision and UI/UX design principle walkthroughs.
Google Drive	Documentation Storage	Cloud storage for internship reports, notes, research summaries, and project documentation shared with supervisors.

6. Key Observations

The activities of Week-02 generated a number of significant observations that will inform the design and development of the Pneumonia Detection System going forward:

- Frontend preparation is not a preliminary formality but a genuine value-generating activity. The time invested in React revision and UI research this week will pay dividends in the quality and speed of the implementation sprint.
- Color psychology is an underappreciated discipline in software engineering education. The research revealed that color choices have measurable, evidence-based impacts on user trust, cognitive load, and emotional state — factors that are magnified in healthcare contexts where users are often anxious or time-pressured.
- React's component-based architecture is particularly well-suited to healthcare applications that need to scale and adapt over time. The ability to compose interfaces from small, independently testable units reduces technical debt and makes feature additions — such as new result visualisation panels — straightforward.
- Accessibility is not optional in medical software. The WCAG guidelines exist precisely to ensure that healthcare tools can be used by the broadest possible population, including users with visual impairments or motor difficulties.
- Documentation is a professional discipline that requires as much care as coding. Writing the Week-01 report clarified my own understanding of the work completed and produced a reference artifact that will be valuable throughout the project.
- Research-driven design produces better outcomes than intuition-driven design. Every color and layout decision made for the Pneumonia Detection System this week has a documented rationale grounded in peer-reviewed research or established design principles.

7. Challenges Faced

Week-02, while primarily research and documentation oriented, presented a set of genuine challenges:

- Understanding healthcare-specific UI requirements required going beyond general UX principles to consider the unique psychological context of medical users. Balancing information density — clinicians need detail, patients need simplicity — was a non-trivial design problem.
- Selecting a suitable color palette from the research findings was more nuanced than expected. Multiple shades of blue and teal are clinically appropriate, but choosing the specific tones that work harmoniously together required iterative comparison and reference to color accessibility checkers.
- Interpreting and synthesising multiple research sources — including academic papers with dense statistical content — demanded careful reading and judgment about which findings were most directly applicable to the project context.
- Balancing aesthetics and functionality in the UI planning was a recurring tension. Design solutions that look visually impressive sometimes introduce complexity that compromises usability, particularly on mobile devices.
- Organising the Week-01 technical documentation into a coherent narrative structure required going back and filling gaps in my understanding of certain technical steps, as the original notes were sometimes brief.

8. Learning Outcomes

Week-02 generated a rich set of learning outcomes spanning technical skills, design knowledge, and professional practices:

- Improved React development skills: I am now confident in building functional components with hooks, managing state, handling events, and structuring a React project for the Pneumonia Detection System.
- Deeper understanding of component-based architecture: I understand how to decompose a complex healthcare interface into a hierarchy of reusable, testable components.
- Enhanced knowledge of UI/UX principles: I can now apply user-centered design, visual hierarchy, accessibility guidelines, and responsive design principles to real frontend work.
- Understanding of healthcare application design: I have developed a specific appreciation for the emotional and cognitive dimensions of design in medical contexts — insights that go well beyond standard web development practice.
- Knowledge of color psychology: I can now make evidence-based color decisions and explain them with reference to peer-reviewed research.
- Improved documentation and reporting skills: Writing the Week-01 report has raised my standards for technical writing and reinforced the habit of documenting not just what was done, but why, and what was learned.
- Stronger research and analytical abilities: Engaging with academic papers and synthesising findings into practical design decisions has sharpened my ability to read and apply research.
- Better project planning capabilities: The review meeting and planning session on Day 1 gave me practical experience of how structured retrospectives and forward planning sessions work in a professional internship environment.

9. Conclusion

Week-02 of the Summer Internship at Akshar AI was a thoughtfully structured preparation week that, while not producing lines of production code, laid essential intellectual and design foundations for the frontend implementation work ahead. The week demonstrated that in software projects — and especially in healthcare software projects — the quality of outcomes in implementation phases is directly determined by the quality of thinking and research that precedes them.

The project review on Day 1 confirmed that the Week-01 EDA and React visualisation deliverables were well received and that the project is on track. The React revision on Day 2 restored and strengthened my confidence in the technology stack that will power the Pneumonia Detection System's user interface. The color psychology research on Day 3 produced a documented, research-backed color palette that will guide every visual design decision. The documentation work on Day 4 produced a thorough, well-structured Week-01 report. The UI/UX research on Day 5 resulted in detailed wireframe concepts for the homepage and authentication system. And the consolidation session on Day 6 synthesised all of this into a clear readiness picture for the implementation sprint.

From a personal and professional development perspective, this week challenged me to engage with fields that sit at the intersection of computer science, psychology, and design — areas that are not always prominent in a data engineering curriculum but are profoundly important in applied AI product development. I leave Week-02 with a greater appreciation for the human dimensions of software systems, and with a strong conviction that the Pneumonia Detection System will be not only technically sound but also genuinely useful and accessible to the people it is designed to serve.

10. References

React and Frontend Development

- [1] Facebook Inc. (2023). React — A JavaScript Library for Building User Interfaces. <https://react.dev/>
- [2] Traversy Media. (2023). Learn React JS — Full Beginner's Tutorial & Practice Projects [YouTube Video]. YouTube. <https://www.youtube.com/>
- [3] MDN Web Docs. (2024). Introduction to the DOM. Mozilla Developer Network. <https://developer.mozilla.org/>
- [4] Hooks Reference — React Documentation. (2023). <https://react.dev/reference/react>

Color Psychology Research

- [5] Elliot, A. J., & Maier, M. A. (2014). Color and Psychological Functioning: A Review of Theoretical and Empirical Work. *Frontiers in Psychology*, 5, 368. <https://doi.org/10.3389/fpsyg.2014.00368>
- [6] Lin, H., et al. (2013). The Effect of Ambient Illumination and Text Color on Visual Fatigue During Screen Reading. *Perceptual and Motor Skills*, 116(3), 782-797.
- [7] Beginning Graphic Design: Color [YouTube Video]. GCFGGlobal. YouTube. <https://www.youtube.com/>
- [8] Astuti, I. (2020). Color Theory for Noobs | Beginner Guide [YouTube Video]. YouTube. <https://www.youtube.com/>

Healthcare UI/UX Design

- [9] Nielsen, J. (2020). 10 Usability Heuristics for User Interface Design. Nielsen Norman Group. <https://www.nngroup.com/articles/ten-usability-heuristics/>
- [10] World Wide Web Consortium (W3C). (2018). Web Content Accessibility Guidelines (WCAG) 2.1. <https://www.w3.org/TR/WCAG21/>
- [11] Chuah, S. H., & Yu, J. (2021). The Future of Service: The Power of Emotion in Human-Robot Interaction. *Journal of Retailing and Consumer Services*, 61, 102551.
- [12] ISO 9241-210. (2019). Ergonomics of Human-System Interaction — Part 210: Human-Centred Design for Interactive Systems. International Organization for Standardization.

Dataset and Project Resources

- [13] Kaggle. (2018). Chest X-Ray Images (Pneumonia). <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>
- [14] Akshar AI Ltd. (2026). System Requirement Proposal — Chest X-Ray Pneumonia Detection and Analysis System Using Deep Learning. Internal Document.
- [15] GitHub. Learn-React Repository. <https://github.com/>